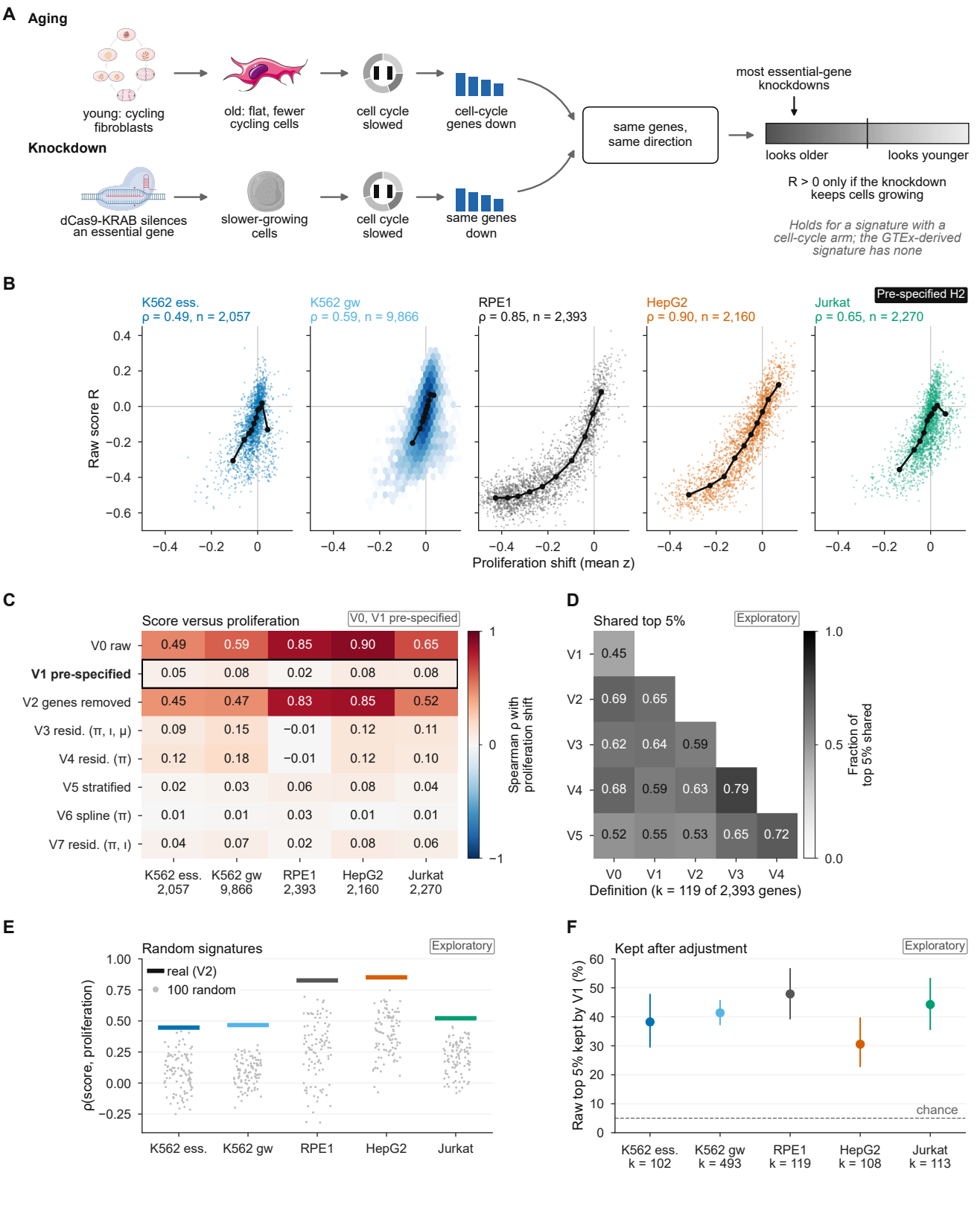

A
Aging
young: cycling fibroblasts
old: flat, fewer cycling cells
cell cycle slowed
cell-cycle genes down
Knockdown
dCas9-KRAB silences an essential gene
slower-growing cells
cell cycle slowed
same genes down
same genes, same direction
most essential-gene knockdowns
looks older
looks younger
R > 0 only if the knockdown keeps cells growing
Holds for a signature with a cell-cycle arm; the GTEx-derived signature has none
B
Pre-specified H2
K562 ess.
ρ = 0.49, n = 2,057
K562 gw
ρ = 0.59, n = 9,866
RPE1
ρ = 0.85, n = 2,393
HepG2
ρ = 0.90, n = 2,160
Jurkat
ρ = 0.65, n = 2,270
Raw score R
Proliferation shift (mean z)
C
Score versus proliferation
V0, V1 pre-specified
V0 raw
V1 pre-specified
V2 genes removed
V3 resid. (π, ı, μ)
V4 resid. (π)
V5 stratified
V6 spline (π)
V7 resid. (π, ı)
0.49 0.59 0.85 0.90 0.65
0.05 0.08 0.02 0.08 0.08
0.45 0.47 0.83 0.85 0.52
0.09 0.15 −0.01 0.12 0.11
0.12 0.18 −0.01 0.12 0.10
0.02 0.03 0.06 0.08 0.04
0.01 0.01 0.03 0.01 0.01
0.04 0.07 0.02 0.08 0.06
K562 ess. 2,057
K562 gw 9,866
RPE1 2,393
HepG2 2,160
Jurkat 2,270
Spearman ρ with proliferation shift
D
Shared top 5%
Exploratory
V1 0.45
V2 0.69 0.65
V3 0.62 0.64 0.59
V4 0.68 0.59 0.63 0.79
V5 0.52 0.55 0.53 0.65 0.72
V0 V1 V2 V3 V4
Definition (k = 119 of 2,393 genes)
Fraction of top 5% shared
E
Random signatures
Exploratory
real (V2)
100 random
ρ(score, proliferation)
K562 ess.
K562 gw
RPE1
HepG2
Jurkat
F
Kept after adjustment
Exploratory
Raw top 5% kept by V1 (%)
chance
K562 ess. k = 102
K562 gw k = 493
RPE1 k = 119
HepG2 k = 108
Jurkat k = 113